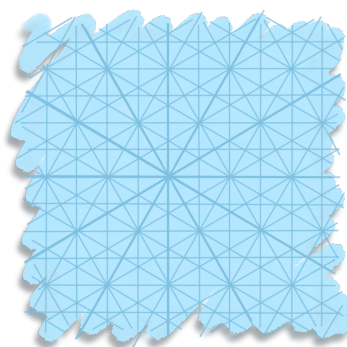
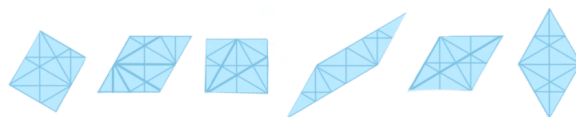


A gist for everyone

In the process of renovating your university's mathematics department, you find yourself – as one does every so often – describing to the architect the pattern that will tile the central floor.



Of course, describing the entire infinite pattern is unnecessary. The architect simply needs the description of a single tile, the “heart” of the pattern, which they may repeat in every direction. For instance, any of the tiles below will suffice, and there are many more such.



The mathematically curious reader may wonder how much freedom they have in choosing this “heart”, and in the process of cataloguing such tiles, convince themselves that any two such can be linked by a sequence of “cut-and-paste” operations as shown.



What is the complete list of hearts associated to this pattern? Which of these tiles are linked by a single cut-and-paste? What is the list of “bricks” – smallest triangles – that makes up any heart?

The present work addresses questions of the above fashion. Tools for probing geometric and algebraic structures (such as those occurring naturally in some physical theories that describe the shape of the universe) often leave us with a kaleidoscopic version of the data we seek. That is to say, we only see an infinitely tiled mosaic in which the true geometry forms one repeating unit.

But, as above, there are many possible geometries from which the same infinite tiling we see could have originated. This thesis studies the structure and symmetries of the “tiled” data – presented in algebraic form as a *triangulated category* – by classifying *hearts of t -structures* and *brick objects* inside it.